

TARGET-DATA ACCESS AND AGGREGATION EFFECTS IN CROSS-SUBJECT EEG EVALUATION

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ABSTRACT

Cross-subject EEG studies give held-out target-subject data different roles across protocols. An accuracy gap measures the performance gain attributable to using those target-subject trials only when other evaluation conditions are aligned; otherwise it may reflect differences in partitioning, sample exposure, model selection, or scoring. We partition each held-out subject's trials into available and reserved halves, score all conditions on the same reserved trials, and cross Euclidean Alignment with supervised training on the available half. The source split, per-epoch sample budget, early-stopping rule, target sampling rate, and realized indices are fixed. Across BCI Competition IV 2a, BNCI2014-002, and PhysionetMI, both procedures improve performance, but their ordering reverses across datasets. A no-retraining regrouping control shows that the standard deviation read as between-subject heterogeneity also changes with trials per group. Thus, target-data value and between-subject spread depend on dataset and evaluation design. Our code is available at <https://github.com/yo666666yo/target-data-access-eeeg>.

Index Terms— Brain-computer interfaces, cross-subject evaluation, Euclidean alignment, calibration data

1. INTRODUCTION

Cross-subject EEG decoders are compared across papers whose evaluation protocols use the target subject's data differently. Subject-independent networks use none of it [1]; Euclidean Alignment and the Riemannian transforms of the same family use the unlabelled signals to align feature distributions [2, 3, 4]; calibration-block finetuning consumes a labelled block outright [5]. The labels attached to these protocols (zero-calibration, unsupervised transfer, subject-adaptive) are coarse, and the accuracy differences between them are routinely read as the value of the extra access.

Partitioning and reporting choices are known to move BCI comparisons [6, 7], and MOABB [8, 9] standardizes the interface but not the accounting: two pipelines can share a loader and a split while giving the target subject different influence over training, normalization and checkpoint choice. Alignment is filed under zero-calibration even though it reads every one of the evaluated subject's trials, so a protocol name records only what a method may label, leaving what it may see unstated. Compact convolutional references [10, 11] are evaluated under all of these protocols. Leakage drives irreproducible findings [12, 13] and small-sample neuroimaging yields error bars wider than reported [14, 15, 16], but that work is diagnostic, whereas choosing between protocols also requires measuring what a procedure is worth once everything else is matched.

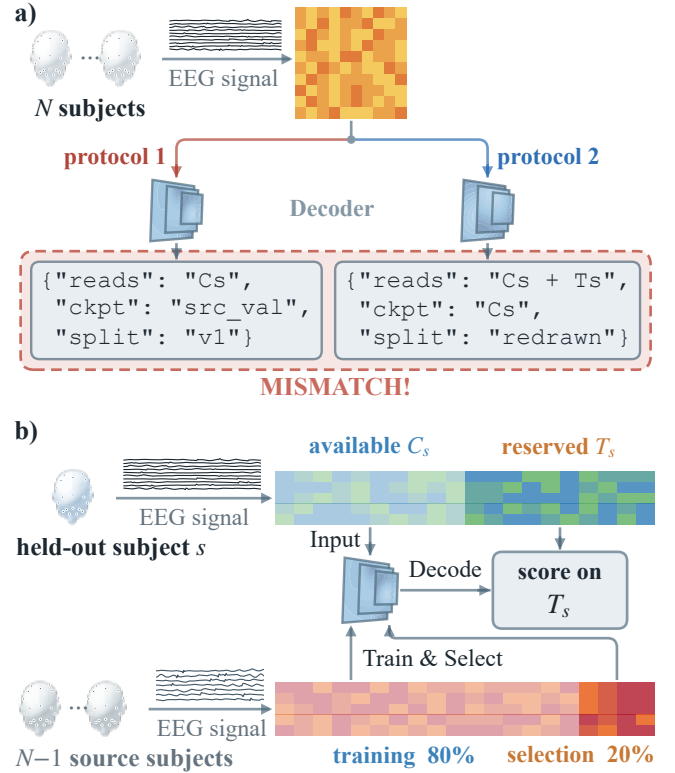


Fig. 1. (a) Two protocols start from the same trials and the same decoder and differ only in how they use them: which target trials they read (C_s , T_s : C_s and T_s of panel b), where the checkpoint is chosen (`src_val`: the source subjects' 20% selection split) and which source split is used (`v1`: the one split every condition shares; `redrawn`: a fresh draw), so their accuracy gap measures more than the target data. (b) The controlled pipeline: the held-out subject s is divided once into an available half C_s , all a condition may use, and a reserved half T_s , all it is scored on; the $N-1$ source subjects give one fixed 80/20 split for training and checkpoint selection. Heads rendered from the Lee Perry-Smith scan (Infinite Realities, CC BY 3.0).

Matching everything else is what published comparisons rarely can do. Adding a calibration block lengthens an epoch, so the adapted model also gets more optimizer steps; the target's share of a batch is whatever the cohort size implies; and if the block supplies the

early-stopping signal, model selection has changed too. None of this is visible in a results table (Fig. 1a), and each can change the effect being claimed. Our own earlier work tracked those quantities rather than the procedure; Section 2.2 reports what replaying it reveals.

This paper narrows the problem to something that can be measured cleanly. On a fixed target test set, we measure what each way of using the target subject’s available trials buys, once the source partition, the per-epoch sample budget, the early-stopping rule and the target sampling rate are held constant. Each held-out subject’s trials are divided once into an available half and a reserved half on which every condition is scored (Fig. 1b). The available half is used in one of four ways: not at all, by Euclidean Alignment, by supervised training, or by both. This crosses two procedures that overlap in what they consume: supervised training reads the available trials’ signals as well as their labels. The grid therefore measures what each procedure adds on top of the other, and leaves the value of signals against labels open.

The same accounting issue arises when the result is summarized. A cross-subject paper reports a spread as well as a mean, and the spread is read as between-subject heterogeneity. That holds only if the aggregation unit is the subject. We test it with a control that retrains nothing: the same per-trial predictions are aggregated by true subject and by random groups of the same sizes, so any difference is attributable to the grouping alone.

We contribute a controlled 2×2 of alignment against target supervision on three motor-imagery datasets, its controls checked against the realized indices; a fixed-prediction regrouping benchmark, with a matching variance estimate, for the aggregation unit’s share of a between-subject spread; and an audit of the same comparison run without these controls. The procedures are standard; the controls, and an account of which of them cannot hold together, are new.

2. METHOD

2.1. Conditions

We evaluate leave-one-subject-out on the three motor-imagery datasets of Table 1. Each held-out subject’s trials are divided once, stratified by class under a per-subject seed, into an available half C_s and a reserved half T_s (Fig. 1b); that 50/50 division is the declared target-data budget, no condition can change it, and T_s is only ever scored.

Figure 2 shows the design. Both procedures read C_s : supervised training on it consumes those trials’ signals as well as their labels. Each axis of the grid names a procedure applied to those trials, and both axes have the same access to them.

Alignment uses signals and no labels. Euclidean Alignment [2] whitens a subject’s trials by $\bar{R}^{-1/2}$, where $\bar{R}$ is the mean over that subject’s trials of the sample spatial covariance $XX^\top/T$, computed on the band-passed epochs with no shrinkage and inverted through an eigendecomposition whose eigenvalues are floored at 10^{-10} to keep a rank-deficient epoch from blowing up. Source subjects are whitened by a reference estimated from all of their trials, which is legitimate because all of them are available. The held-out subject’s reference is estimated from C_s alone and then frozen; T_s is transformed by it but never contributes to it. The switch also changes how every source subject is preprocessed, so the measured effect is that of the whole alignment pipeline. Target supervision adds C_s to the training set with its labels attached.

The four conditions are neither procedure (SRC), alignment only (EA), supervision only (SUP), and both (EA+SUP). Two further arms sit off the grid, each changing a third thing: SEL picks the check-

point on C_s rather than the source validation set, leaving training set, initialization and trajectory unchanged but swapping a selection set of very different size; POOL adds C_s to the same scaler’s fit set. T_s enters no scaler.

2.2. Controlled variables

Three quantities are pinned exactly:

- the source partition: the stratified 80/20 split supplying training and selection data is derived from the outer pool alone and is the same index set in all four conditions, and validation is source-only throughout, so target trials never reach the selection set;
- the epoch: every condition draws, with replacement, as many samples per epoch as the source training set holds (3686, 1664 and 7486);
- the scored object: T_s is identical across conditions, trial for trial.

A fourth, the target sampling rate, is fixed in expectation: a plain shuffled loader over the same pool would give the target 7.2%, 4.6% and 0.6% of a batch, a 12-fold range set by cohort size. The supervised arms therefore use a weighted sampler with a declared expected share of 0.10, realized at 0.096–0.109.

Two quantities this does not equalize, and we report both. Because the epoch is a fixed total, a supervised arm draws 10% fewer source samples per epoch than SRC; and because each arm early-stops on its own curve, realized updates differ: taking fold-averaged totals per arm and dividing the spread by the smallest, by 10% on 2a, 3% on PhysionetMI and 23% on BNCI2014-002. Exposure is the sharper problem: a declared share w implies $wS/|C_s|$ expected draws of each target trial per epoch, which is 1.3, 2.1 and 16.6 here, since $|C_s|$ differs 6.4-fold. Matching repetition instead would need shares of 0.156, 0.096 and 0.012; matching both would need source exposure to vary 13-fold. Any such comparison fixes two of these three and lets the third move, which is why we vary $|C_s|$ directly in Section 3.

An earlier version of this study enforced none of these controls, and replaying it gives the pattern of Fig. 1a: a supervised arm on a re-drawn source partition with target trials in its validation set and batch shares of 6.1, 3.8 and 0.5%, which its cross-dataset ordering followed; a selection arm with 25% more source trials than its reference; and a whitener fitted on $C_s \cup T_s$. None shows in a results table, so a script checks the controls against the realized indices before any run.

2.3. Aggregation unit

A cross-subject spread is read as between-subject heterogeneity, which requires the aggregation unit to be the subject. We test that with a control that retrains nothing: the per-trial predictions on T_s are held fixed and only the grouping changes, first by true subject, then into random groups of the same sizes, 2000 times, on SRC and SUP. Trial-weighted accuracy is identical either way, and subjects

Table 1. The three motor-imagery datasets. Trials per half is the declared target-data budget: each held-out subject is cut once into an available half C_s and a reserved test half T_s of that size. Chance is given in parentheses; it differs across the datasets, so accuracies are not comparable between them.

	Subjects	Channels	Classes	Trials/half
BCI IV 2a [17]	9	22	4 (0.25)	288
BNCI2014-002 [18]	14	15	2 (0.50)	80
PhysionetMI [19, 20]	105	64	4 (0.25)	45

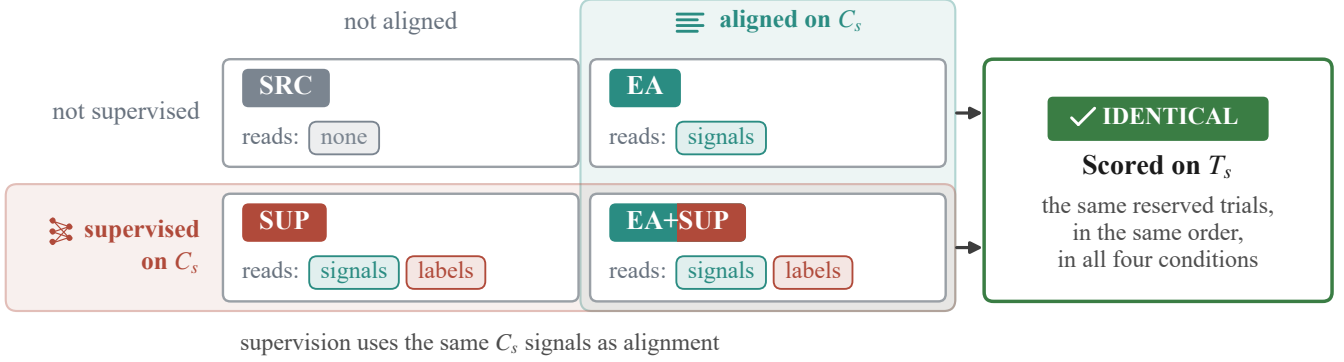


Fig. 2. The four conditions as a 2×2 over the two procedures, with what each reads from the held-out subject’s available half C_s . Supervised training reads those trials’ signals as well as their labels, so the procedures overlap. Everything else is held fixed, as Section 2.2 sets out, and all four are scored on the same reserved trials in the same order.

contribute equal trial counts on 2a and BNCI2014-002 and near-equal ones on PhysionetMI (45 for 104 subjects, 44 for one), so the subject macro-average and the trial-weighted average coincide up to that single trial and any difference in spread belongs to the grouping alone. The null is not a new phenomenon: drawing m of M correctness records without replacement gives group means of variance $(1 - m/M)S^2/m$, which our resampled null reproduces. What it adds is how much of a reported spread that accounts for, where m ranges from 45 to 288.

2.4. Statistics

Each subject contributes one number per condition, averaged over three runs. Those runs vary the target split and the initialization together, so they estimate neither alone; within a run all conditions share both, which is what the pairing requires. Differences are paired within subject and resampled over subjects, with one set of bootstrap indices drawn per dataset and reused for every condition (10^4 replicates, percentile intervals [21]). p -values come from a two-sided sign-flip permutation test [22] over the five conditions of a dataset, Holm-corrected [23] within that family, with a paired t -test as a sensitivity check. That test saturates on a small cohort: on nine subjects it cannot return less than $2/2^9 = 0.0039$, so we report the number of subjects helped beside it. Spread comparisons are ratios of standard deviations, so every factor quoted below is on that scale.

Because leave-one-subject-out folds share most of their training pool, the subjects are not independent replicates [24], and with nine of them on 2a the reference distribution is coarse. We therefore read a Holm-adjusted p below 0.05 as a difference resolved by this repetition structure (these subjects, these runs, this training policy) and not as inference about a subject population, and we use that as the single rule for which differences we state. Intervals give magnitude only: they carry no multiplicity adjustment, and in a Gaussian simulation at $n = 9$ our percentile intervals covered 89% rather than 95%, so an interval excluding zero beside a non-significant adjusted p is not evidence of an effect.

3. RESULTS

3.1. Setup

BCI Competition IV 2a [17] is read from the competition archives, band-passed 8–32 Hz over the four seconds after the cue, with EOG

dropped by the official channel contract and sessions T and E concatenated in that order. BNCI2014-002 [18] is right hand versus feet. PhysionetMI [19, 20] drops its rest class, and subjects 88, 89, 92 and 100 are excluded for non-standard sampling. The latter two are loaded through MOABB [8] with its motor-imagery paradigm at its default 8–32 Hz, each over its own declared trial interval.

The decoder is our own reimplement of EEGNet [10], with $F_1 = 8$ temporal filters, depth multiplier $D = 2$, $F_2 = 16$ pointwise filters and a temporal kernel of 64 samples. It omits the max-norm constraints of the original, one of which braindecode’s implementation [11] applies, so Section 3.3 repeats the whole grid with that decoder swapped in. Training uses Adam [25] (10^{-3} , weight decay 10^{-4}), batch 64, ≤ 50 epochs, step decay ($\times 0.5$ every 15) with patience 10 and class-balanced loss, with features standardized by one scaler fitted on source training trials.

3.2. Condition effects

Table 2 gives the source-only references and the effects. Chance differs across the three datasets, so the reference accuracies are not comparable to one another; every claim below is a within-subject difference against an arm’s own reference. Supervision rejects on all three datasets and alignment on two of three, with effect sizes that do not transfer across them, and the ordering between the two reverses.

Alignment is worth +1.58 percentage points (pp) on 2a, where it does not survive correction (7 of 9 subjects, Holm $p = 0.18$), against +8.07 and +9.12 pp on BNCI2014-002 and PhysionetMI, where it does (12/14 and 93/105, $p \leq 0.005$). Target supervision runs the other way: +9.45 pp on 2a, helping every subject, falling to +6.52 and +4.89 pp on the larger cohorts. Comparing the two directly is exploratory, prompted by this ordering rather than specified in advance. On that comparison supervision leads alignment by +7.87 pp [+4.95, +11.03] on 2a, is indistinguishable from it on BNCI2014-002 (−1.55 [−4.46, +0.89]), and trails it by −4.23 pp [−5.84, −2.64] on PhysionetMI. Both endpoints reject under the decision rule (Holm-adjusted within these three tests, $p = 0.008$ and $p < 0.001$), so the observed contrasts do not support a common ordering across datasets. Being exploratory, the reversal stands as an observation in these runs and awaits a pre-specified test.

The calibration budget is confounded with the dataset, $|C_s|$ differing 6.4-fold, so we re-ran the arms that consume it with $|C_s|$ capped by a nested stratified subset and T_s untouched. Shrinking it moves the effect monotonically but only a little: at a common budget of 45 trials the three datasets read +5.88, −2.62 and −4.23 pp, so the

Table 2. Effect of each procedure against its own source-only reference under the controls of Section 2.2, paired within subject. Intervals are unadjusted percentile bootstrap; p is Holm-corrected and followed by the number of subjects helped (Section 2.4). Figures are from our own EEGNet implementation; Section 3.3 gives the braindecode counterpart.

		BCI IV 2a	BNCI2014-002	PhysionetMI
Source-only acc		0.450	0.687	0.444
EA	Δ acc (pp)	+1.58 [+0.17, +2.98]	+8.07 [+4.37, +11.96]	+9.12 [+7.40, +10.81]
	p (# helped)	0.176 (7/9)	0.005 (12/14)	< 0.001 (93/105)
SUP	Δ acc (pp)	+9.45 [+5.88, +12.81]	+6.52 [+3.18, +10.27]	+4.89 [+3.41, +6.41]
	p (# helped)	0.019 (9/9)	0.005 (12/14)	< 0.001 (72/105)
EA+SUP	Δ acc (pp)	+13.66 [+9.53, +17.44]	+11.13 [+6.34, +16.16]	+12.68 [+11.04, +14.35]
	p (# helped)	0.019 (9/9)	0.002 (13/14)	< 0.001 (99/105)

ordering holds and the budget accounts for only about 2 of the 12 pp separating 2a from PhysionetMI. Within 2a the lead of supervision falls from +7.87 pp at 288 available trials to +7.29 at 144 and +5.88 at 45, as each trial goes from being drawn 1.3 times per epoch to 8.4; within BNCI2014-002 it moves from -1.55 to -2.62 , and only at the smaller budget does it resolve ($p = 0.049$ against $p = 0.33$). Budget does move the contrast, and in the direction that shrinks it, but by less within a dataset than separates one dataset from another.

Whether the two procedures compose additively is undetermined. Write the interaction as the gain alignment contributes on top of supervision, less the gain it contributes on its own,

$$\Delta_{\text{int}} = (\text{EA} + \text{SUP} - \text{SUP}) - (\text{EA} - \text{SRC}). \quad (1)$$

Its estimate is +2.62, -3.45 and -1.33 pp on the three datasets, and none of the three is resolved under the decision rule of Section 2.4. The sign of Δ_{int} is consistent across runs within a dataset but not across datasets, so it may itself be dataset-dependent. We therefore report neither evidence for additivity nor evidence against it.

3.3. Control checks

Two things could make these effects artefacts of our setup rather than of the protocol. The random split interleaves the halves in time, so they share drift and attention state; BCI IV 2a’s two sessions let the same 288/288 budget be partitioned chronologically, and over five runs the effects change little: the lead of supervision over alignment is +7.75 pp [+5.54, +10.29] under the chronological partition against +7.87 pp under the random one, which makes that contrast cross-session. And the decoder is our own reimplementation, so we repeated the grid with braindecode’s EEGNetv4 (0.8.1, matched hyper-parameters) swapped in: source-only accuracy moves by -0.81 , $+0.27$ and -0.55 pp, so the low reference is not attributable to our implementation alone, and the reversal reproduces, at +8.27 pp on 2a and -4.65 pp on PhysionetMI. EA and SUP reproduce wherever they reject; SEL on BNCI2014-002 does not. These runs and the capped-budget ones carry their own source snapshots.

3.4. Aggregation effects

Holding the predictions fixed and changing only the grouping, the resampled null matches the finite-population prediction to within 0.0013 and tracks the trials per group, though it also depends on the pooled accuracy and the group count. With 288 test trials each on 2a it is 0.029 against 0.177 by true subject, a factor of 6.2, falling to 2.4 with 80 trials and 1.8 with 45; the supervised arm behaves likewise (6.99, 2.96, 2.23), though not at the same size, so the gap is not independent of training.

That null serves as a benchmark: shuffling removes the subject structure along with the trial noise, so it cannot decompose the spread. Estimating the two components separately instead, under a conditionally binomial model, puts finite trials at 2%, 17% and 29% of the observed between-subject variance and leaves standard deviations of 0.174, 0.111 and 0.113 against raw values of 0.177, 0.122 and 0.134; the estimator, its assumptions and the per-run terms are in the repository. Raw figures rank PhysionetMI as the more heterogeneous cohort and corrected ones do not, though that gap is unresolved (+0.002 [-0.031 , +0.024]). The control itself establishes only that spread exceeds the benchmark everywhere ($p < 0.001$).

3.5. Off-grid arms

Choosing the checkpoint on C_s can cost accuracy: SEL is -1.69 pp [-2.80 , -0.55] on PhysionetMI, helping 37 of 105 subjects, surviving correction ($p = 0.009$) and reproducing under the reference decoder (-1.81). That is where the selection set is smallest, 45 trials against the 1872 it replaces. Its positive values elsewhere do not reproduce, so we claim only this direction. No POOL arm rejections.

4. DISCUSSION

Some readings these numbers do not support. The reference accuracies sit below those usually quoted, but standardization, budget, schedule, alignment and a reference decoder each failed to account for the gap, so the discrepancy cannot be attributed to our implementation alone. A large alignment gain is likewise no evidence of a violation: admissibility turns on whether the claimed deployment permits reading the target’s unlabelled signals. Nor is the supervised effect the value of labels. A fixed batch share replays each PhysionetMI target trial 16.6 times per epoch against 1.3 on 2a, which made overfit to a small, repeatedly drawn C_s a candidate explanation for the reversal; capping $|C_s|$ makes that explanation insufficient on its own. The remaining difference is dataset-dependent in these runs, but we do not resolve its cause further. SEL’s harm is a separate route: it never trains on C_s , only picks a checkpoint there, so what is at stake is selection noise on 45 validation trials rather than training overfit, a distinction these runs do not resolve.

Limitations are two decoders, one alignment method, one sampling rate, and one 50/50 budget; the chronological split was available only on 2a. Section 2.4 states the decision rule and the intervals’ limits.

COMPLIANCE WITH ETHICAL STANDARDS

This work is a secondary analysis of three public motor-imagery EEG corpora: BCI Competition IV dataset 2a, BNCI2014-002 and the PhysioNet EEG Motor Movement/Imagery database. Each was released by its originators for open research use, with participant consent and ethical approval obtained by the original investigators. No new data were recorded and no human participants were involved, so no additional approval was required. The authors declare no conflict of interest.

5. REFERENCES

- [1] O-Yeon Kwon, Min-Ho Lee, Cuntai Guan, and Seong-Whan Lee, "Subject-independent brain-computer interfaces based on deep convolutional neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 31, no. 10, pp. 3839–3852, 2020.
- [2] He He and Dongrui Wu, "Transfer learning for brain-computer interfaces: A Euclidean space data alignment approach," *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 2, pp. 399–410, 2020.
- [3] Paolo Zanini, Marco Congedo, Christian Jutten, Salem Said, and Yannick Berthoumieu, "Transfer learning: A Riemannian geometry framework with applications to brain-computer interfaces," *IEEE Transactions on Biomedical Engineering*, vol. 65, no. 5, pp. 1107–1116, 2018.
- [4] Pedro Luiz Coelho Rodrigues, Christian Jutten, and Marco Congedo, "Riemannian Procrustes analysis: Transfer learning for brain-computer interfaces," *IEEE Transactions on Biomedical Engineering*, vol. 66, no. 8, pp. 2390–2401, 2019.
- [5] Dongrui Wu, Yifan Xu, and Bao-Liang Lu, "Transfer learning for EEG-based brain-computer interfaces: A review of progress made since 2016," *IEEE Transactions on Cognitive and Developmental Systems*, vol. 14, no. 1, pp. 4–19, 2022.
- [6] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, "A review of classification algorithms for EEG-based brain-computer interfaces: A 10 year update," *Journal of Neural Engineering*, vol. 15, no. 3, pp. 031005, 2018.
- [7] Yannick Roy, Hubert Banville, Isabela Albuquerque, Alexandre Gramfort, Tiago H. Falk, and Jocelyn Faubert, "Deep learning-based electroencephalography analysis: A systematic review," *Journal of Neural Engineering*, vol. 16, no. 5, pp. 051001, 2019.
- [8] Vinay Jayaram and Alexandre Barachant, "MOABB: Trustworthy algorithm benchmarking for BCIs," *Journal of Neural Engineering*, vol. 15, no. 6, pp. 066011, 2018.
- [9] Sylvain Chevallier, Igor Carrara, Bruno Aristimunha, et al., "The largest EEG-based BCI reproducibility study for open science: the MOABB benchmark," *arXiv preprint arXiv:2404.15319*, 2024.
- [10] Vernon J. Lawhern, Amelia J. Solon, Nicholas R. Waytowich, Stephen M. Gordon, Chou P. Hung, and Brent J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, pp. 056013, 2018.
- [11] Robin Tibor Schirrmeister, Jost Tobias Springenberg, Lukas Dominique Josef Fiederer, Martin Glasstetter, Katharina Eggensperger, Michael Tangermann, Frank Hutter, Wolfram Burgard, and Tonio Ball, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.
- [12] Sayash Kapoor and Arvind Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, pp. 100804, 2023.
- [13] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Pascal Vincent, et al., "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems*, 2021.
- [14] Gaël Varoquaux, "Cross-validation failure: Small sample sizes lead to large error bars," *NeuroImage*, vol. 180, pp. 68–77, 2018.
- [15] Andrius Vabalas, Emma Gowen, Ellen Poliakoff, and Alexander J. Casson, "Machine learning algorithm validation with a limited sample size," *PLOS ONE*, vol. 14, no. 11, pp. e0224365, 2019.
- [16] Russell A. Poldrack, Grace Huckins, and Gaël Varoquaux, "Establishment of best practices for evidence for prediction: A review," *JAMA Psychiatry*, vol. 77, no. 5, pp. 534–540, 2020.
- [17] Michael Tangermann, Klaus-Robert Müller, Ad Aertsen, Niels Birbaumer, Christoph Braun, Clemens Brunner, Robert Leeb, Carsten Mehring, Kai J. Miller, Gernot R. Müller-Putz, Guido Nolte, Gert Pfurtscheller, Hubert Preissl, Gerwin Schalk, Alois Schlögl, Carmen Vidaurre, Stephan Waldert, and Benjamin Blankertz, "Review of the BCI competition IV," *Frontiers in Neuroscience*, vol. 6, pp. 55, 2012.
- [18] David Steyerl, Reinhold Scherer, Josef Faller, and Gernot R. Müller-Putz, "Random forests in non-invasive sensorimotor rhythm brain-computer interfaces: a practical and convenient non-linear classifier," *Biomedical Engineering / Biomedizinische Technik*, vol. 61, no. 1, pp. 77–86, 2016.
- [19] Gerwin Schalk, Dennis J. McFarland, Thilo Hinterberger, Niels Birbaumer, and Jonathan R. Wolpaw, "BCI2000: A general-purpose brain-computer interface (BCI) system," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 1034–1043, 2004.
- [20] Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley, "PhysioBank, PhysioToolkit, and PhysioNet," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [21] Bradley Efron and Robert J. Tibshirani, *An Introduction to the Bootstrap*, Chapman and Hall/CRC, 1994.
- [22] Anderson M. Winkler, Gerard R. Ridgway, Matthew A. Webster, Stephen M. Smith, and Thomas E. Nichols, "Permutation inference for the general linear model," *NeuroImage*, vol. 92, pp. 381–397, 2014.
- [23] Sture Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [24] Claude Nadeau and Yoshua Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.
- [25] Diederik P. Kingma and Jimmy Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations*, 2015.